Process Mathematics: Classical Analysis Without Completed Infinity

Formal Verification in Rocq with 21,603 Theorems

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Abstract

We develop a formal framework for classical mathematical analysis in which infinity is treated as a property of processes rather than completed objects. The central construction is `RealProcess` $:= \mathbb{N} \to \mathbb{Q}$—a function from natural numbers to rationals—which replaces the real number line $\mathbb{R}$.

Within this framework, formalized in the Rocq proof assistant (v9.0.1), we establish:

1. Fifteen theorems from Wiedijk's "100 theorems" benchmark, plus seven additional results of comparable significance (IVT, EVT, Bolzano–Weierstrass, Heine–Borel, FTC, MVT, Taylor, Picard–Lindelöf, Stone–Weierstrass, and others), all without the Axiom of Infinity or Axiom of Choice.
2. Process completeness: every Cauchy sequence of processes converges stagewise to a diagonal process, with an explicit metric and convergence proof. We analyze the relationship to classical completeness and state precisely what the Cauchy-completeness bridge does and does not provide.
3. Proof-theoretic analysis: ordinals to ε_0 with well-foundedness proved as a theorem, and an explicit discussion of how P4 relates to Church's Thesis and to the strength of the ambient proof assistant.

The formalization comprises 21,603 machine-verified theorems across 1,483 files with 0 Admitted and 2 axioms (classical logic and constructive witness extraction).

Companion papers: The physics applications (E/R/R framework, Weinberg angle, Born rule) and the data compression pipeline are treated in separate papers [6, 7].

Keywords: process mathematics, constructive analysis, formal verification, Rocq, Cauchy sequences, P4, ordinals, Wiedijk benchmark

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1 Introduction

1.1 The Problem

Standard mathematical analysis rests on the real number line $\mathbb{R}$, conceived as a completed, uncountable set. This construction requires the Axiom of Infinity (to assert the existence of $\mathbb{N}$ as a completed set) and typically the Axiom of Choice. The Theory of Systems [1] proposes an alternative: derive mathematics from the principle $A =$ exists, with infinity treated as a property of *processes* (Principle P4), not of completed objects.

1.2 Scope of This Paper

This paper establishes the *mathematical* foundations: the universal type `RealProcess`, process completeness, proof-theoretic strength, and classical analysis theorems. We do not treat the physics applications or the data compression pipeline; these are the subject of companion papers [6, 7].

1.3 Contributions

1. **P4 as ontological position** (§2). We present P4 transparently: it is equivalent to a finitist ontology augmented with classical logic. We analyze its relationship to Church's Thesis and state precisely what it entails for the function space $\mathbb{N} \to \mathbb{N}$.

2. **Process completeness** (§3). We define a metric on `RealProcess`, construct the diagonal limit, prove stagewise completeness, and give an honest analysis of what this does and does not recover from classical completeness.

3. **Proof-theoretic calibration** (§4). We reach ε_0, prove well-foundedness, and discuss the relationship between P4's claimed restrictions and Rocq's actual strength.

4. **Classical analysis without Infinity** (§5). Nine theorems in the unified E/R/R pattern: process + Cauchy + bound = existence.

5. **Wiedijk benchmark** (§6). Fifteen of the 100 theorems, all without the Axiom of Infinity.

2 The Universal Type and P4

2.1 RealProcess

Definition 2.1 (RealProcess)**.** A *real process* is a function $R : \mathbb{N} \to \mathbb{Q}$. We write `RealProcess` $:= \mathbb{N} \to \mathbb{Q}$.

Definition 2.2 (Cauchy condition)**.** A real process R is *Cauchy* if for every $\varepsilon \in \mathbb{Q}$, $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$: $|R(m) - R(n)| < \varepsilon$.

In Rocq:

```
Definition RealProcess := nat -> Q.

Definition is_Cauchy (R : RealProcess) : Prop :=
  forall eps : Q, 0 < eps ->
    exists N : nat, forall m n : nat,
      (N <= m)%nat -> (N <= n)%nat ->
      Qabs (R m - R n) < eps.
```

Under P4, the Cauchy process *is* the real number. There is no separate limit object.

2.2 P4 as Ontological Position

We state P4 precisely and transparently.

> **P4 (Finite Actuality):** At any stage, finitely many objects are actual. The type `nat` is an inductive type (a process of succession), not a completed set. Every function $\mathbb{N} \to \mathbb{N}$ that exists is given by a finite description.

Relationship to Church's Thesis. The last clause—every function $\mathbb{N} \to \mathbb{N}$ is finitely described—is equivalent to the *ontological* form of Church's Thesis: every total function on natural numbers is computable. We adopt this as an explicit philosophical position, not as a derived consequence. The collapse of Π^1_1 quantification to arithmetic (§4.4) is then a direct corollary, not a surprise.

What P4 entails:

- The Axiom of Infinity (a completed $\mathbb{N}$ exists) is rejected.
- The full Axiom of Choice (selection from uncountable families) is rejected.
- Countable choice (selection from a process of finite stages) is compatible.
- The law of excluded middle (L3) is retained—P4 is not constructivism.

What P4 does *not* entail by itself. P4 alone does not determine the proof-theoretic strength of the system. That depends on which inductive types and recursion principles are admitted. See §4.5 for a discussion.

2.3 Incompatibility with Completed Infinity

Definition 2.3. $\text{CompletedInfSet}(S) :\Leftrightarrow \forall n : \mathbb{N},\ S(n)$.

Definition 2.4. $\text{P4_bounded}(\text{actual}) :\Leftrightarrow \forall \text{stage},\ \exists \text{bound},\ \forall n,\ \text{actual}(\text{stage}, n) \Rightarrow n \leq \text{bound}$.

Theorem 2.5. *If S is a completed infinite set,* actual *is P4-bounded, and every member of S is actual at stage 0, then $\bot$.*

Proof. P4 gives bound B at stage 0. $S(B{+}1)$ holds. The bridge makes $B{+}1$ actual. But $B{+}1 > B$, contradicting the bound. □

Remark 2.6. The force of this theorem depends on the bridge assumption: "every member of S is actual at stage 0." This is not a logical necessity but an ontological commitment: if a completed set exists, its members are simultaneously actual. Under P4, this commitment is precisely what is rejected. The theorem shows that P4-boundedness and the bridge are *jointly* inconsistent with completed infinity—not that completed infinity is "refuted" in any absolute sense.

Rocq: `P4CompletedInfinity.v`, theorem `completed_inf_contradicts_P4`, Qed.

3 Process Completeness

This section addresses the central objection: "Without $\mathbb{R}$, Cauchy sequences may not converge." We define a metric, construct limits, and analyze precisely what is and is not recovered.

3.1 Stage Distance

Definition 3.1 (Stage distance)**.** For processes R, S : `RealProcess` and truncation level N:

$$d_N(R, S) = \sum_{k=0}^{N-1} |R(k) - S(k)|.$$

Properties verified in Rocq (`ProcessMetricComplete.v`):

$$d_N(R, R) = 0, \tag{1}$$
$$d_N(R, S) \geq 0, \tag{2}$$
$$d_N(R, S) \leq d_{N+1}(R, S). \tag{3}$$

Under P4, the "full" distance $d(R, S) = \lim_{N\to\infty} d_N(R, S)$ does not exist as a completed object. The sequence $\{d_N(R, S)\}$ is itself a process over $\mathbb{Q}$.

3.2 Process-Cauchy and Diagonal Limit

Definition 3.2 (Process-Cauchy)**.** A sequence $\mathrm{seq} : \mathbb{N} \to$ `RealProcess` is *process-Cauchy* if for every truncation N and every $\varepsilon > 0$, there exists M such that for all $i, j \geq M$: $d_N(\mathrm{seq}(i), \mathrm{seq}(j)) < \varepsilon$.

Definition 3.3 (Diagonal process)**.** $R_\infty(k) = \mathrm{seq}(k)(k)$.

Theorem 3.4 (Stagewise completeness)**.** *If* seq *is process-Cauchy, then for every stage k and $\varepsilon > 0$, there exists M such that for all $i \geq M$ with $i \geq k$:*

$$|\mathrm{seq}(i)(k) - R_\infty(k)| < \varepsilon.$$

Proof. Process-Cauchy at $N = k+1$ gives M_k with $|\mathrm{seq}(i)(k) - \mathrm{seq}(j)(k)| < \varepsilon$ for $i, j \geq M_k$. Take $j = k$ (requiring $k \geq M_k$, so set $M = \max(M_k, k)$): $|\mathrm{seq}(i)(k) - \mathrm{seq}(k)(k)| = |\mathrm{seq}(i)(k) - R_\infty(k)| < \varepsilon$. □

Rocq: `ProcessMetricComplete.v`, theorem `process_completeness`, QED.

3.3 Relationship to Classical Completeness

Stagewise completeness says: for each *fixed* k, the k-th components converge. This is *not* the same as convergence in a completed metric space. We now state precisely what the bridge to classical completeness provides.

What `ProcessBridge.v` establishes:

1. A Cauchy process $R : \mathbb{N} \to \mathbb{Q}$ can be packaged as a classical Cauchy sequence of rationals (a `CauchySeq` record in Rocq).
2. If classical $\mathbb{R}$ is available (e.g., via Rocq's `Reals` library), then every `CauchySeq` has a limit in $\mathbb{R}$.
3. The composition gives: every Cauchy process determines a real number.

What it does *not* establish:

1. It does not construct $\mathbb{R}$ within P4. Step 2 imports the classical reals.
2. It does not prove that the process metric (stagewise distance) is complete in the classical sense.
3. It does not show that every real number arises from a Cauchy process (only the converse direction is trivial).

Honest assessment

Under P4, we have *stagewise completeness*: every Cauchy sequence of processes converges componentwise to a diagonal process. This suffices for all nine classical analysis theorems in §5: each constructs a Cauchy process and bounds its rate, never needing a completed limit object.

Classical completeness—the existence of a limit *point* in a complete metric space—is available via the bridge, but the bridge imports $\mathbb{R}$ from outside the P4 framework. Under P4 alone, the process *is* the real number, and stagewise completeness is the operative notion.

3.4 Consequences for the Function Space

If every function $\mathbb{N} \to \mathbb{N}$ is computable (the ontological commitment of P4, see §2.2), then:

- The "reals" (Cauchy processes over $\mathbb{Q}$) are *countable*—only computable Cauchy sequences exist.
- Continuous-but-nowhere-differentiable functions exist only if computably given.
- Large parts of descriptive set theory (Borel hierarchy beyond Σ^0_1, analytic sets) are modified.

These are genuine mathematical losses. We argue (§7) that they do not affect any theorem used in science or engineering, but we do not claim they are negligible from a mathematical standpoint.

4 Proof-Theoretic Strength

4.1 Ordinals to ε_0

```
Inductive Ord : Set :=
  | OZero : Ord
  | OSucc : Ord -> Ord
  | OLim  : (nat -> Ord) -> Ord.
```

Concrete ordinals defined: $\omega =$ `OLim(nat_to_ord)`, ω^ω, and $\varepsilon_0 =$ `OLim(omega_tower)`. Full arithmetic (addition, multiplication, exponentiation) verified.

4.2 Well-Foundedness as Theorem

Theorem 4.1. *`wf_ord_lt` : `well_founded ord_lt`.*

Proved by structural induction on `Ord` with an accessibility argument (helper `acc_succ`), requiring no additional axioms.

Rocq: `TransfiniteInduction.v`, 12 Qed.

4.3 ATR_0 as Definition

Under P4, transfinite recursion along `Ord` is a `Fixpoint`—accepted by Rocq's termination checker because recursion is structural. The "axiom" ATR_0 becomes a function definition.

Rocq: `P4_Eliminates_ATR.v`, 15 Qed.

4.4 Π^1_1 and Church's Thesis

Under P4's ontological commitment (every function $\mathbb{N} \to \mathbb{N}$ is finitely described), second-order quantification collapses:

$$\forall f : \mathbb{N} \to \mathbb{N}.\ \varphi(f, n) \quad \equiv \quad \forall c : \mathbb{N}.\ \varphi(\mathtt{eval}(c), n).$$

We emphasize: this is a *corollary* of the P4 ontology (which is equivalent to the ontological form of Church's Thesis), not an independent derivation. The result is unsurprising given the premise; its value lies in making the consequence explicit.

Rocq: `P4_Eliminates_Pi11.v`, 15 Qed.

4.5 P4 vs. Rocq's Strength

A methodological issue deserves transparency. Rocq's Calculus of Inductive Constructions (CIC) has proof-theoretic strength far beyond ε_0—it can define ordinals past Γ_0 and beyond. Our formalization lives *within* Rocq but does not *use* its full strength.

What we claim and what we don't

We claim: constructions up to ε_0 *suffice* for classical analysis (IVT, EVT, BW, FTC, MVT, Taylor, Picard, and 15 Wiedijk theorems). This is demonstrated by the 21,603 Qed.
We do not claim: ε_0 is the *maximum* that P4 allows. A formal characterization of "P4-admissible inductive types" within CIC is an open problem. Our position is: the interesting question is not what P4 *prohibits* but what suffices.

4.6 Position in the Hierarchy

System	L3	Ax. Inf.	AC	Ordinal	Coverage
PRA	varies	no	no	ω^ω	primitive recursive
PA	yes	no	no	ε_0	first-order arithmetic
ACA_0	yes	yes	no	ε_0	predicative analysis
ZFC	yes	yes	yes	inaccessible	all standard math
ToS	**yes**	**no**	**no**	$\boldsymbol{\varepsilon_0}^*$	**classical analysis**

Table 1: Proof-theoretic hierarchy. *Constructions to ε_0 demonstrated; upper bound not formally established (see §4.5).

5 Classical Theorems as Process Constructions

Every classical existence theorem has the same structure: construct a process over $\mathbb{Q}$, show it is Cauchy with computable rate, verify the property at every stage.

5.1 Intermediate Value Theorem

Theorem 5.1 (IVT, Process Version)**.** *Let $f : \mathbb{Q} \to \mathbb{Q}$ be uniformly continuous on $[a, b]$ with $f(a) < 0 < f(b)$. Then there exists a Cauchy process $c : \mathbb{N} \to \mathbb{Q}$ with $c(n) \in [a, b]$ and $|f(c(n))| \to 0$.*

Construction. Bisection: $c(n)$ is the midpoint of a nested interval halving at each step. Rate: $|c(m) - c(n)| \leq (b - a)/2^{\min(m,n)}$. Uses L3 for the sign test.
Rocq: `IVT_ERR.v`, 24 Qed.

5.2 Extreme Value Theorem

Grid refinement: $x^*(n) = \arg\max_k f(a + k(b-a)/2^n)$ on a 2^n-point grid. The supremum process is monotonically non-decreasing. When multiple points achieve the maximum, L5-Resolution selects the leftmost.
Rocq: `EVT_ERR.v`, 36 Qed.

5.3 Bolzano–Weierstrass

Nested interval bisection: at each step, at least one half contains infinitely many terms (pigeonhole + L3). Rate: interval width $(b-a)/2^n$.
Rocq: `BolzanoWeierstrass.v`, 27 Qed.

5.4 Fundamental Theorem of Calculus

Riemann sums on 2^n-point grids form a Cauchy process. Lipschitz bound: $|F(n+1) - F(n)| \leq L(b-a)^2/2^{n+1}$.

Rocq: `analysis/FTC.v`, 29 Qed.

5.5 Further Theorems

Mean Value Theorem (`MeanValueTheorem.v`, 18 Qed), Taylor's Theorem (`TaylorSeries.v`, 18 Qed), Picard–Lindelöf (`PicardLindelof.v`, 24 Qed), Heine–Borel (`HeineBorelComplete.v`, 30 Qed), Monotone Convergence (`MonotoneConvergence.v`, 15 Qed).

5.6 The Unified Pattern

Theorem	Elements	Roles	Rules	Rate
IVT	midpoints	Cauchy, bounded	bisection	$1/2^n$
EVT	grid maximizers	monotone	grid refinement	$\delta(2^{-n})$
BW	intervals	shrinking	pigeonhole (L3)	$1/2^n$
Heine–Borel	grid points	covering	Lebesgue number	finite
FTC	Riemann sums	Cauchy	Lipschitz bound	$1/2^n$
MVT	critical pts	extremal	IVT + EVT	$\delta(2^{-n})$
Taylor	partial sums	bounded error	remainder	$1/n!$
Picard	iterates y_n	Cauchy	contraction	$(Lt)^n/n!$
Monotone CV	bounded seq	converging	supremum	$1/n$

Table 2: Nine process theorems. Each: process over $\mathbb{Q}$, Cauchy condition, explicit rate.

6 Wiedijk's 100 Theorems

#	Theorem	Rocq File	Qed
1	$\sqrt{2}$ irrational	`Sqrt2Irrational.v`	14
25	Schroeder–Bernstein	`SchroederBernstein_ERR.v`	16
34	Harmonic series diverges	`HarmonicDiverges.v`	21
35	Taylor's theorem	`TaylorSeries.v`	18
39	Fibonacci / golden ratio	3 files	34
58	Binomial coefficients	`Combinatorics.v`	26
63	Cantor's theorem	2 files	215
64	L'Hôpital's rule	`LHopital.v`	25
69	GCD	`GCD.v`	23
74	Cauchy–Schwarz	`CauchySchwarz.v`	18
75	Bolzano–Weierstrass	`BolzanoWeierstrass.v`	27
78	Mean value theorem	`MeanValueTheorem.v`	18
79	Intermediate value thm	`IVT_ERR.v`	24
80	Fundamental thm calculus	`FTC.v`	29
86	Lebesgue measure	`LebesgueMeasure.v`	23
Total from Wiedijk's 100:			**531**

Table 3: Fifteen theorems from Wiedijk's list, all proved over $\mathbb{Q}$ without the Axiom of Infinity.

Seven additional results (EVT, Heine–Borel, Stone–Weierstrass, Picard–Lindelöf, Implicit Function, Banach Fixed Point, Monotone Convergence) bring the total to 22 theorems and 703 Qed.

7 What Survives, What Is Lost

7.1 What Survives

All theorems in Tables 2 and 3, plus the broader formalization (21,603 QED covering algebra, topology, measure theory, ODEs, spectral theory, and lattice gauge theory). Every computable mathematical statement that holds in ZFC also holds under P4.

7.2 What Is Lost—Honestly

P4 with the Church's Thesis commitment entails:

1. **The reals are countable.** Only computable Cauchy sequences exist. The classical uncountability of $\mathbb{R}$ is replaced by the statement: no computable enumeration captures all computable Cauchy sequences (a weaker but still meaningful diagonal result, formalized in `ShrinkingIntervals_ERR.v`).

2. **Non-measurable sets do not exist.** Every definable set of reals is measurable. Banach–Tarski is dissolved. This is widely regarded as a feature, not a bug.

3. **Descriptive set theory is modified.** The Borel hierarchy, analytic sets, and the theory of definability change character when only computable objects exist.

4. **Full Kruskal's tree theorem is unavailable.** We prove it for specific families (chains, forks, bounded depth) but the full minimal bad sequence argument requires Π^1_2 strength, which exceeds our demonstrated ε_0.

5. **Continuous-but-nowhere-differentiable functions** exist only if given by a computable rule. The "generic" continuous function of classical analysis—which exists by a Baire category argument over non-computable objects—does not exist under P4.

7.3 Assessment

No physical prediction, no engineering algorithm, and no computational procedure depends on the existence of non-computable functions or non-measurable sets. The mathematical losses are real but confined to areas of pure mathematics that concern the behavior of non-constructive objects. Whether this is an acceptable tradeoff is a philosophical judgment, not a mathematical one.

8 Formalization Statistics

Domain	Files	QED
Foundation (laws, principles, E/R/R)	117	1,589
Process framework (Phases 0–10)	297	4,180
Standard library	627	7,510
Gauge theory	114	2,177
Classical analysis	34	590
Other (zeta, physics, experimental, ...)	294	5,557
Total	**1,483**	**21,603**

Table 4: Formalization statistics. 0 Admitted, 2 axioms (`classic`, `L4_witness`).

Project	Theorems	Wiedijk	Logic	Ax. ∞
Lean `mathlib`	200,000+	100	CIC	yes
Isabelle AFP	150,000+	76	HOL	yes
Mizar MML	60,000+	69	set theory	yes
ToS	**21,603**	**15**	**CIC + classic**	**no**

Our contribution is not size but foundations: we demonstrate that classical analysis does not require completed infinities.

9 Conclusion

Classical analysis works without completed infinities. The type `RealProcess` $= \mathbb{N} \to \mathbb{Q}$, combined with a finitist ontology (P4) and classical logic (L3), suffices for 22 classical analysis theorems, 15 from Wiedijk's benchmark, with process completeness via diagonal construction.

The framework occupies a unique position: classical logic without the Axiom of Infinity, reaching ε_0 in proof-theoretic strength. Its losses are real (countable reals, no non-measurable sets, modified descriptive set theory) but do not affect computable mathematics.

The companion papers [6, 7] demonstrate that the same framework supports physics (Weinberg angle, Born rule, gauge theory) and engineering (300× IoT compression).

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